

RC BOAT REPORT

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1. Abstract

This project presents the design and implementation of a remotely controlled (RC) boat using an ESP32 microcontroller. The system is capable of driving two DC motors through an L298N motor driver while being powered by rechargeable lithium-ion batteries.

A voltage regulator (LM2596 buck converter) is used to provide a stable supply voltage for the ESP32. The boat is designed to be controlled wirelessly using a smartphone through the ESP32's built-in communication capabilities.

The project demonstrates the integration of embedded systems, power electronics, motor control, and wireless communication into a practical robotic application.

2. Objectives

The objectives of this project are:

- Design a simple RC boat control system.
- Drive two DC motors independently.
- Build a stable power supply for the electronics.
- Interface the ESP32 with an H-Bridge motor driver.
- Develop embedded software for motor control.
- Implement wireless control using a smartphone.
- Learn practical wiring and debugging of embedded hardware.

3. System Overview

The RC boat consists of four major subsystems:

1. Power Supply
2. ESP32 Control Unit
3. Motor Driver
4. Propulsion System

The ESP32 sends digital control signals to the L298N motor driver, which controls the direction of rotation of the two DC motors.

The motors provide differential steering, allowing the boat to move forward, backward, left, and right without using a servo motor.

4. Components Used

- ESP32 Development Board
- L298N Dual H-Bridge Motor Driver
- LM2596 DC-DC Buck Converter
- Two DC Brushed Motors
- Two 18650 Lithium-Ion Batteries
- Battery Holder
- ON/OFF Switch
- Breadboard
- Jumper Wires
- USB Cable

5. Working Principle

The batteries supply power to the LM2596 buck converter.

The LM2596 regulates the battery voltage and provides a stable voltage for the ESP32.

The ESP32 generates digital output signals that control the L298N motor driver.

The L298N switches the current flowing through the motors, allowing forward and reverse rotation.

By controlling the two motors independently, the boat performs differential steering.

6. Hardware Connections

The power system of the RC boat is supplied by two 18650 lithium-ion batteries connected to an ON/OFF switch. The output of the switch is connected to the input of the LM2596 DC-DC buck converter, while the negative terminal of the battery is connected directly to the negative input of the LM2596.

The output of the LM2596 provides a regulated voltage to both the ESP32 development board and the L298N motor driver. The positive output of the regulator is connected to the VIN pin of the ESP32 as well as the VS power input of the L298N motor driver. The negative output of the LM2596 is connected to the GND pin of the ESP32 and the GND terminal of the L298N, creating a common ground shared by the entire system.

The ESP32 controls the L298N motor driver through four digital GPIO pins. GPIO25 is connected to IN1, GPIO26 is connected to IN2, GPIO27 is connected to IN3, and GPIO14 is connected to IN4. These pins determine the rotation direction of the two DC motors.

The two DC motors are connected directly to the output terminals of the L298N motor driver. The first motor is connected between OUT1 and OUT2, while the second motor is connected between OUT3 and OUT4.

The ENA and ENB enable pins of the L298N are permanently enabled using the onboard jumper caps. This configuration allows both motors to operate at full speed during the initial testing phase without requiring PWM speed control from the ESP32.

Finally, all ground connections of the system, including the ESP32, the LM2596 voltage regulator, the L298N motor driver, and the battery pack, are connected together to establish a common electrical reference. This common ground is essential for reliable communication between the microcontroller and the motor driver and ensures stable operation of the entire RC boat system.

7. Motor Control Logic

Forward

Motor A → Forward
Motor B → Forward

Backward

Motor A → Reverse
Motor B → Reverse

Left

Motor A → Stop
Motor B → Forward

Right

Motor A → Forward
Motor B → Stop

Stop
Both motors OFF

8. Conclusion

This project successfully demonstrated the design and implementation of a wireless RC boat based on the ESP32 microcontroller. The integration of the ESP32, LM2596 power regulator, and L298N motor driver provided a reliable and efficient embedded control platform for two DC propulsion motors. The prototype verified the correct operation of the electrical hardware, motor driver, and embedded software.